

DANIEL B. CARPENTER, MS

Bentonville, AR • dbry1024@gmail.com • daniel-carpenter.github.io/Portfolio/ • linkedin.com/in/dbry1024

EXPERIENCE

Senior FP&A Advisor

November 2022 – Present

Office of Investment Management & Strategic Finance, the Chickasaw Nation

- Researched forecasting methods by developing and deploying Python and R models with LASSO, Elastic Net, LSTM, Naïve Bayes, and kNN, applying feature selection, and evaluating with ROC, RMSE, and cross validation.
- Conducted competitor entry market analysis using SQL and R regression, visualized with ggplot2 and Tableau, and directly presented with Canva to executives, uncovering an ineffective marketing strategy.
- Engineered an automated Python data pipeline with validation checks for a ~\$3B portfolio, improving data quality, governance, and allocation accuracy while reducing manual entry risk.
- Expanded executive access to insights by creating an R Shiny dashboard and Quarto reporting suite, centralizing financial and strategic reporting for the CIO and Chief Executive.

Operational Consultant

January 2021 – November 2022

Office of Strategic Planning, the Chickasaw Nation

- Estimated ~\$8M in revenue uplift by applying optimization modeling in Python (Gurobi) with simulation and feature selection, visualized in Tableau to translate an unstructured retail placement challenge into a variable vendor commission framework.
- Conducted causal econometric analysis in R and SQL that revealed online advertising was not generating incremental revenue, presenting findings in Power BI to inform enterprise marketing strategy.
- Improved customer service while reducing costs by designing optimal workforce scheduling models in Python, R, and SQL, incorporating labor drivers and constraints, with outputs visualized in Tableau for leadership review.
- Discovered \$100M in unanticipated service offerings through post-COVID market analysis in R, delivering data-driven insights that expanded support options for customers.
- Prevented \$12M in projected annual costs by analyzing compensation structures in R and Tableau, presenting polished recommendations in PowerPoint that influenced leadership decisions.

Operational Consultant Intern

June 2018 – December 2020

- Reduced cost of goods sold by 42% for a precision manufacturer by developing inventory optimization models in R and SQL, with results delivered through Tableau dashboards.
- Standardized labor metrics across enterprises by building workforce models in R and Tableau, improving performance comparisons, and reducing staffing inefficiencies.

AREAS OF EXPERTISE & SKILLS

- **Coding and Testing, Data Acquisition and Strategy:** Python, R, SQL, GitHub
- **Analytical Modeling:** Logistic Regression, Decision Trees, Random Forest, Gradient Boosted Machine (GBM), kNN, SVM, Regression and Classification (Naïve Bayes, MLP Classifier, LASSO, Elastic Net, PLS, PCR, MARS), Exploratory Data Analysis (Feature Selection, Normalization, Imputation), Bayes, Simulation, Heuristics, Excel, Power Query
- **Model Deployment and Assessment:** ROC, RMSE, Chi-square, Cross Validation, Kubernetes
- **Data Visualization:** Tableau, Power BI, R Shiny, ggplot2, plotly, matplotlib, seaborn

EDUCATION

Master of Science, Data Science and Analytics

2023

University of Oklahoma, Norman, OK – College of Engineering – GPA: 3.72/4.00

Bachelor of Business Administration, Economics and Finance

2020

University of Oklahoma, Norman, OK – College of Business – GPA: 3.83/4.00 (Special Distinction)

HONORS & ACCOMPLISHMENTS

Distinguished Leadership Award, Children's Miracle Network Hospitals Dance Marathon, May 2020

- Nationally recognized for impact on Children's Hospital funding; selected from three hundred colleges in CMN.